Fiscal Year:	FY 2021	**Task Last Updated:**	FY 11/01/2021
PI Name:	Singh, Nitin Ph.D.		
Project Title:	MANGO: An Interactive Tool for Microbiome Data Analysis and Visualization		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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City:	Pasadena	**State:**	CA
Zip Code:	91109-8001	**Congressional District:**	27
Comments:			
Project Type:	Ground,NASA GeneLab	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research
Start Date:	04/03/2019	**End Date:**	08/02/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 8/2/2021 per F. Hernandez/ARC (Ed., 4/29/2020) NOTE: Period of performance changed to 4/3/2019-4/3/2021 (from 2/1/19-2/28/21) per F. Hernandez/ARC and PI (Ed., 12/13/19) NOTE: End date changed to 2/28/2021 per F. Hernandez/ARC (Ed., 6/12/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Barker, Richard Ph.D. (University of Wisconsin System)		
Grant/Contract No.:	JPL Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	We will generate a computational tool, Microbiome Analysis of NASA GeneLab Omics (MANGO), which will enable the visualization of all the microbial next-generation sequencing (NGS) data that are not yet curated but available on the NASA GeneLab Data System (GLDS). The MANGO tool is a manually curated, custom data visualization software. Currently, the GLDS is a simple repository of "omics" data generated by NASA-funded projects, without any computational tools to analyze these complex datasets. MANGO is a Systems Biology Informatics (SBI) research tool that will extend the GLDS investigations. This tool will also help to develop new computational algorithms and techniques to perform novel informatics research and produce new informatics products that enhance the value of GLDS for all future open science-based investigations. MANGO will overcome current limitations and promote innovations to ease the usability of the GLDS for future open science-based investigations, and it will provide easy-to-interpret insights into the microbiomes of the International Space Station (ISS), revealing potential changes in the microbial community dynamics. MANGO will establish unified processing for ISS genomic data at the GLDS using a custom pipeline to overcome experimental biases across different time point experiments, providing insights across different experiments conducted on the ISS. It will allow easy to-adapt graphical user interface (GUI) based interactive 3D visualizations using JavaScript and D3 libraries in the Qlik platforms (QlikTech International), linking data from studies of various microbiome features such as microbial burden, biofilms, virulence, and antibiotic resistance. This tool will be a modular visualization system, making it easy to integrate different databases like Test of Arabidopsis Space Transcriptome (TOAST), leading to a better understanding of any similarities between microbial and plant adaptation to the spaceflight environment. This will be a first-of-its-kind data analysis tool custom made for GeneLab data to allow not only scientists, but also fellow citizens, to test and interpret biological hypotheses.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	MANGO will be a first-of-its-kind data analysis tool custom made for GeneLab data to allow not only space scientists but also national research institutes of science and medicine, academic research institutions, and private corporations as per GeneLab guidelines to test and interpret biological hypotheses. As a global scientific community comes together to conduct joint research in the fields of biology, genomics, systems biology, and data analysis, NASA GeneLab believes this “open science” approach has the potential to democratize science and produce a source of new knowledge greater than any single institution could deliver. In addition, NASA Space Biology Science Plan 2016-2025 engages the broadest possible community of researchers, industry, and the general public to foster innovation. The MANGO tool development directly responds to both GeneLab “open science” policy and Space Biology Science Plan guidelines.
Task Progress:	Due to the COVID 19 shutdown, the project asked for an extension of 4 months; this was a no-cost extension. Study was on course and slight modifications were made in the period of performance. As per the JPL record, the Period of Performance changed to April 3, 2019 through August 2, 2021. FINAL REPORTING OCTOBER 2021 This software accomplishes the following: Microbiome Analysis of NASA GeneLab Omics (MANGO) enables the visualization of all the microbial next-generation sequencing (NGS) data that are not yet curated but available on the NASA GeneLab Data System (GLDS). The MANGO tool is a manually curated, custom data visualization software. Currently, the GLDS is a simple repository of "omics" data generated by NASA-funded projects with limited computational tools to analyze these complex datasets. MANGO is a Systems Biology Informatics (SBI) research tool that will extend the GLDS investigations. This tool helps to develop new computational algorithms and techniques to perform novel informatics research and produce new informatics products that enhance the value of the GLDS for all future open science-based investigations. What are the unique features of the software? MANGO is a modular visualization system, making it easy to integrate different databases like SEED, KEGG, TOAST etc., leading to a better understanding of any similarities between microbe and host adaptation to the spaceflight environment. MANGO is a first-of-its-kind data analysis tool custom made for GeneLab data to allow not only space scientists but also national research institutes of science and medicine, academic research institutions, and private corporations, as per GeneLab guidelines, to test and interpret biological hypotheses. What improvements have been made over existing similar software applications? MANGO is a first-of-its-kind data analysis tool custom made for GeneLab and space related data analysis. Software Release Version: Mango v.1 MANGO overcomes current limitations and promotes innovations to ease the usability of the GLDS for future open science-based investigations. It provides easy-to-interpret insights into the microbiomes of the ISS, revealing potential changes in the microbial community dynamics. MANGO establishes unified processing for International Space Station (ISS) genomic data deposited at the GLDS, using a custom pipeline to overcome experimental biases across different time point experiments, providing insights across different experiments conducted on the ISS. It allows easy-to-adapt graphical user interface (GUI) based, interactive 3D visualizations using JavaScript and D3 libraries in the Qlik platforms linking data from studies of various microbiome features such as microbial burden, biofilms, virulence, and antibiotic resistance. ANNUAL REPORTING FEBRUARY 2021 The goal of this project is to develop a visualization tool aimed at rapid correlative analysis of all ISS microbiome data available on GeneLab. The MANGO project consists of the following tasks. Task 1. Metadata collection and consolidation of all ISS microbiome NGS data (Jet Propulsion Laboratory-JPL) - Completed Task 2. SPROUT-based processing of NGS data for unbiased downstream processing and curation (JPL) - Completed Task 3. GUI based MANGO platform development using Qlik (University of Wisconsin-Madison-UWM) - In progress The MANGO team in total analyzed 289 samples from the ISS and spacecraft assembly facility. This makes it currently the most extensive metagenomic study associated with any NASA project present in GeneLab. The samples consist of controls and propidium monoazide (PMA) treated, which gives an insight into the viable microbiome population in spacecraft associated surfaces. MANGO gives detailed insight into the two systems where we can see correlative

	conjunctions. The evaluation website is now online. The future plan is to continue refinement of the web interface and error correction. Integrating the different experiments into a correlative analysis for generation of new scientific knowledge, and increasing the return on investment in the GLDS, is also involved. The final product of MANGO will be completely interactive and allow rapid filtering of the GLDS metagenomics to allow rapid exploration of the metadata.
Bibliography Type:	Description: (Last Updated: 10/22/2025)
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